Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

To Eliminate Back Door Clinching Open issue on Scorpio BIW for All Variants.

Mahindra and Mahindra Ltd, Nasik Plant-1, SBS.

	Plan	Actual
Start Date	**17.05.2012**	**17.05.2012**
Completion Date	**20.09.2012**	**20.10.2012**

Current Status:

Completed

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Gajanan Kanade	L8-O	Tool Engg	9604546486	KANADE.GAJANAN@mahindra.com
2	**Mentor**	Suryavanshi R.G	L4-DH	HOD SBS	9881378988	SURYAVANSHI.RAVINDRA@mahindra.com
3	**Member 1**	Vivek Bhat	L7-O	Tool Engg	9922438403	BHAT.VIVEK@mahindra.com
4	**Member 2**	Gangurde Mangesh	L8-O	Mfg QA	9922644399	GANGURDE.MANGESH@mahindra.com
5	**Member 3**	Dyneshwar Vispute	L8-O	Prod.	9923508932	VISPUTE.DYNESHWAR@mahindra.com

1. Select the Theme

Brief description of Problem

What: **Back Door clinching opening concern**

Where: **Stn#30 inside the back door clinching press**

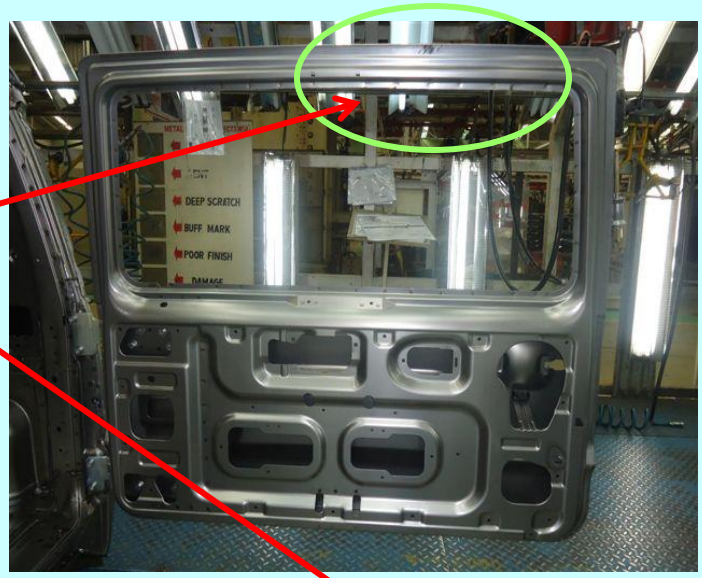
Who: **Concern reported by Buyoff operator**

When: **Daily (1st and 2nd shift)**

How: **Avg. @ 62 incidences per month**

Brief description of Problem

Back Door clinching concern as shown



Cost Of Poor Quality (COPQ)

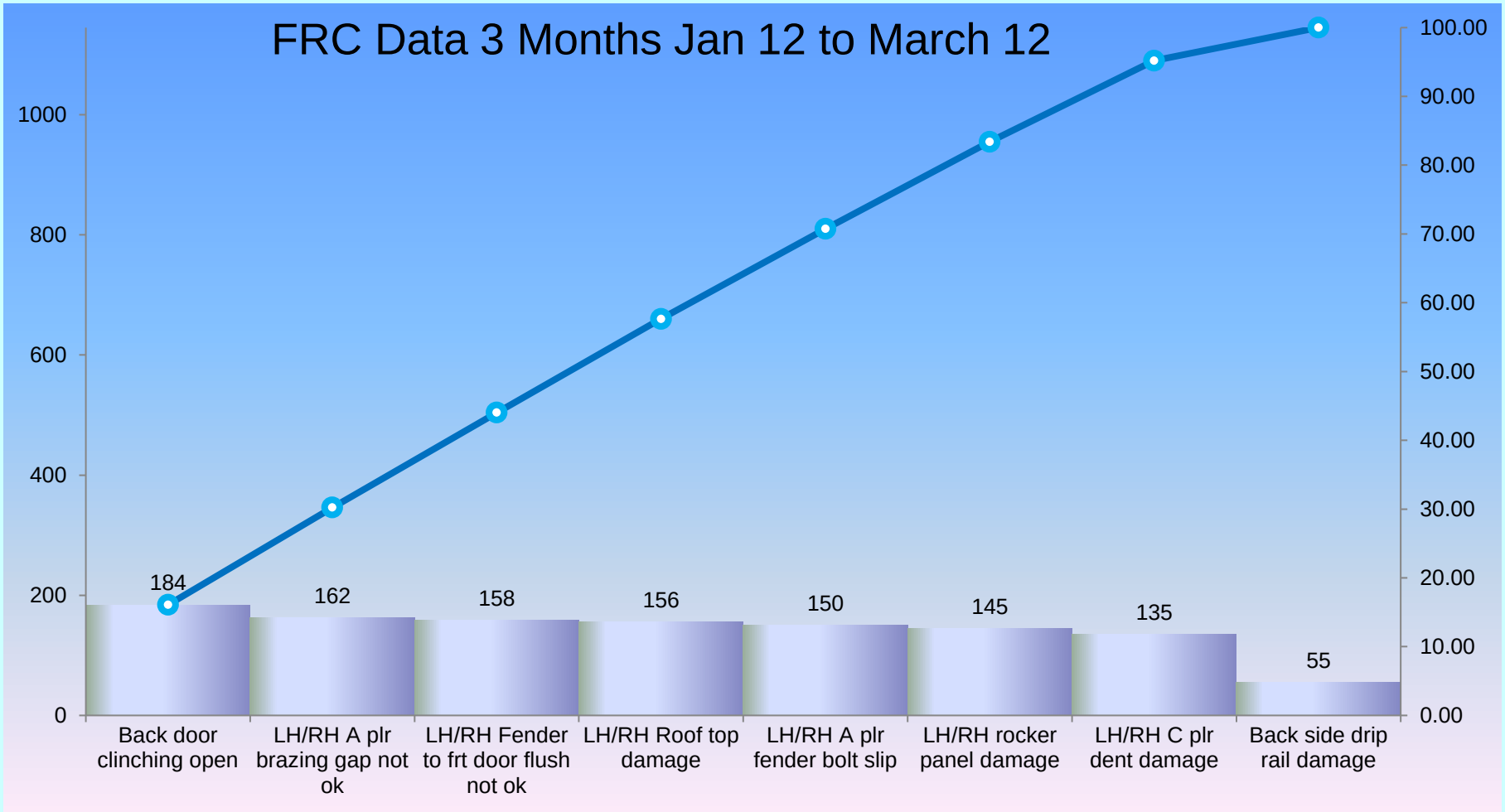
Avg. @ 62 incidences per month

Rework time required for one Door = 20-25 mins

Rework cost for one Door = 27.41 Rs

Total Rework cost for all defective Doors/month is= 27.41*62 Rs ie 1699 Rs

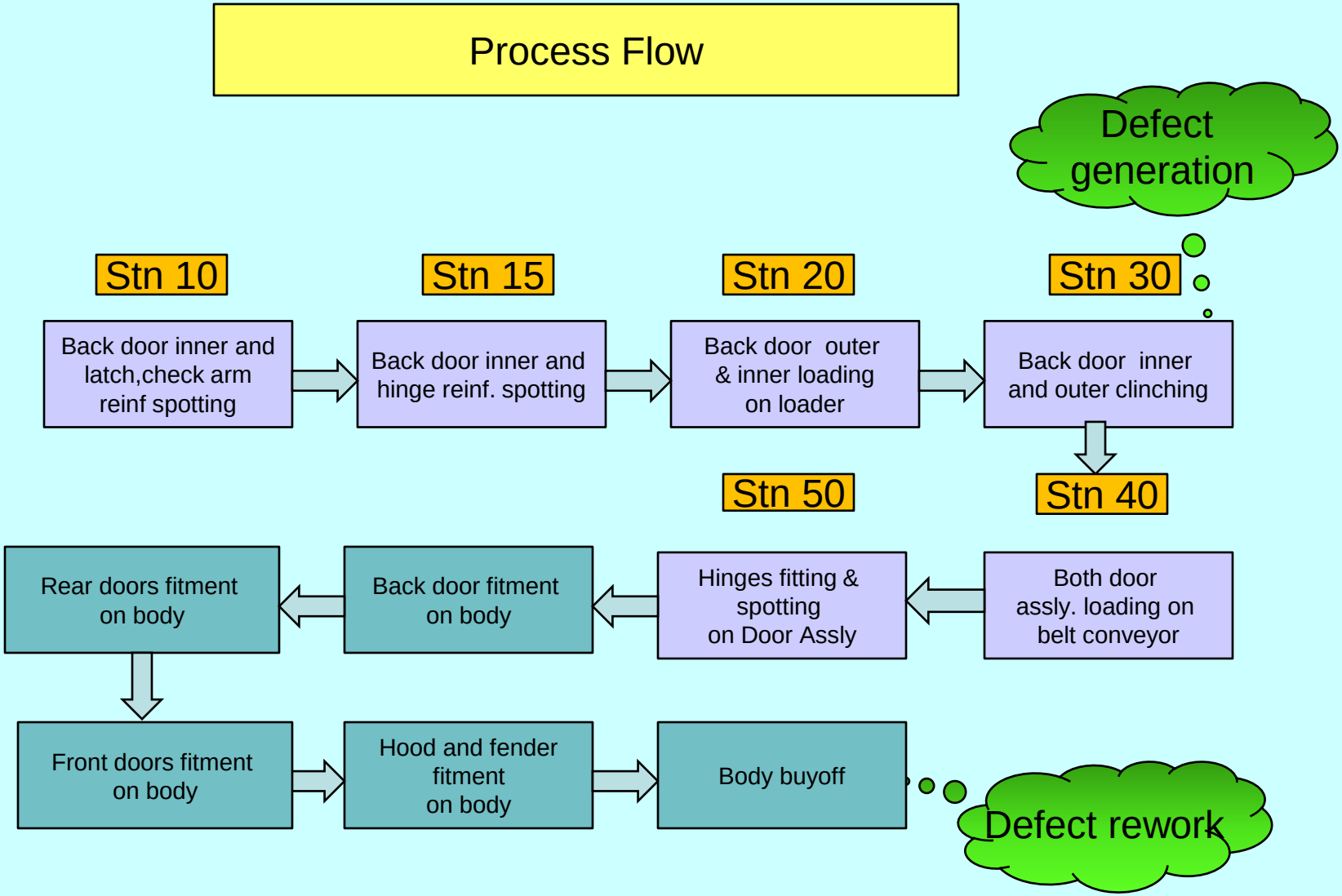
2. Justify the choice (reasons)



2.a Immediate Containment Action

In order to protect our immediate customer ie Paint shop we have started 100% checking at our body buyoff and rework of defective doors

3. Understand the current situation



Process Flow-Back door Clinching

Stn 10

Stn 15

Stn 20

Defect generation

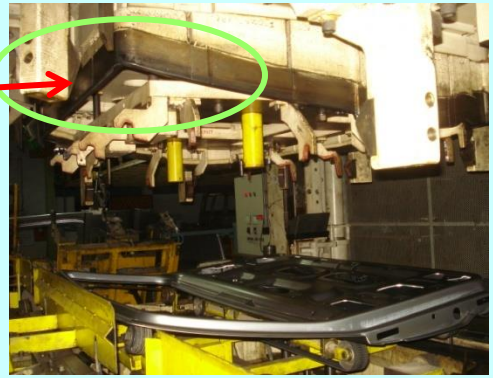
Stn 30

Stn 40

Stn 50

Process Flow-Back door Clinching

Upper Die Hemming blocks performs the operation of

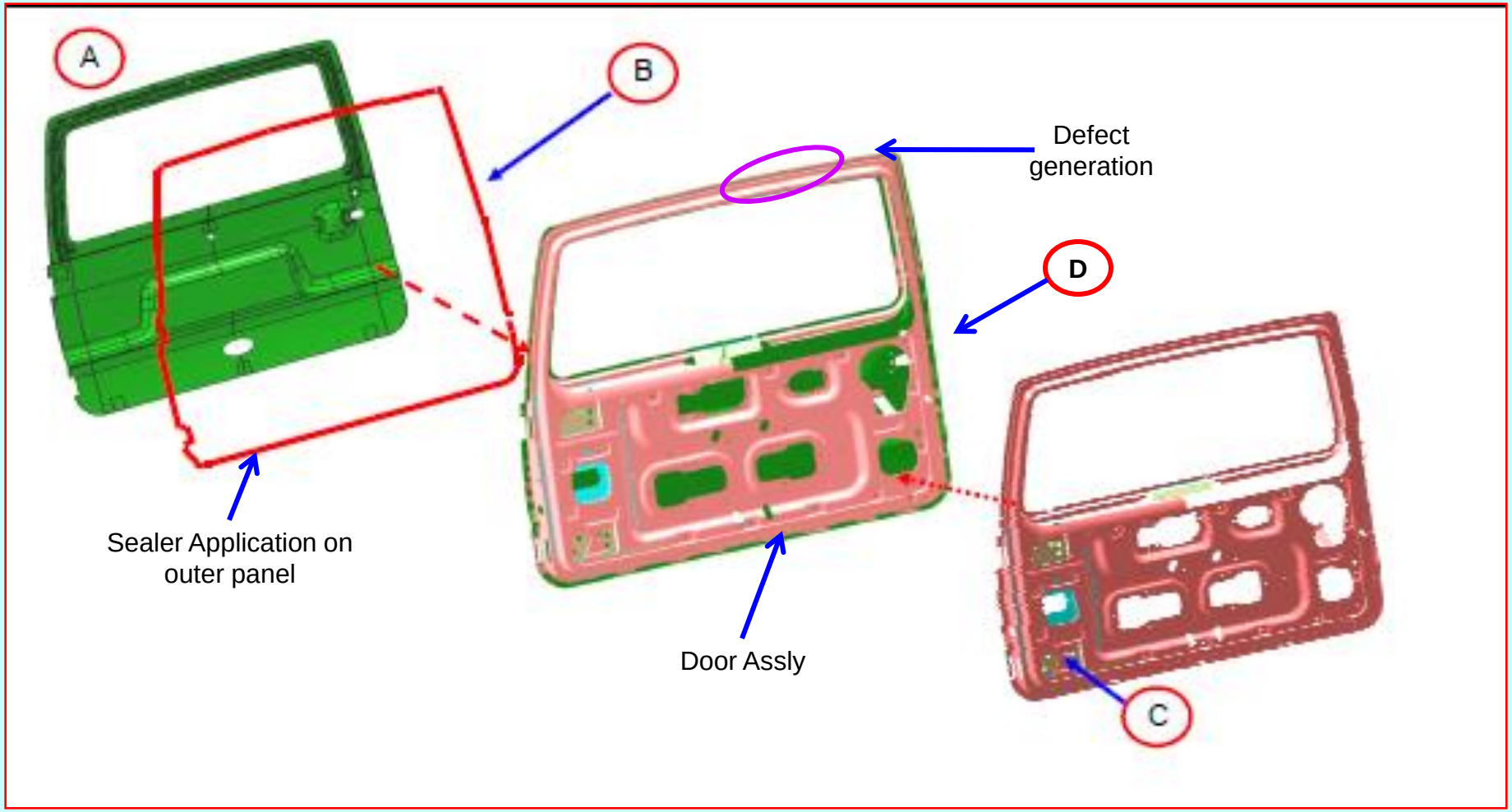


Press Ram is at at the BDC in this operation clinching process completes

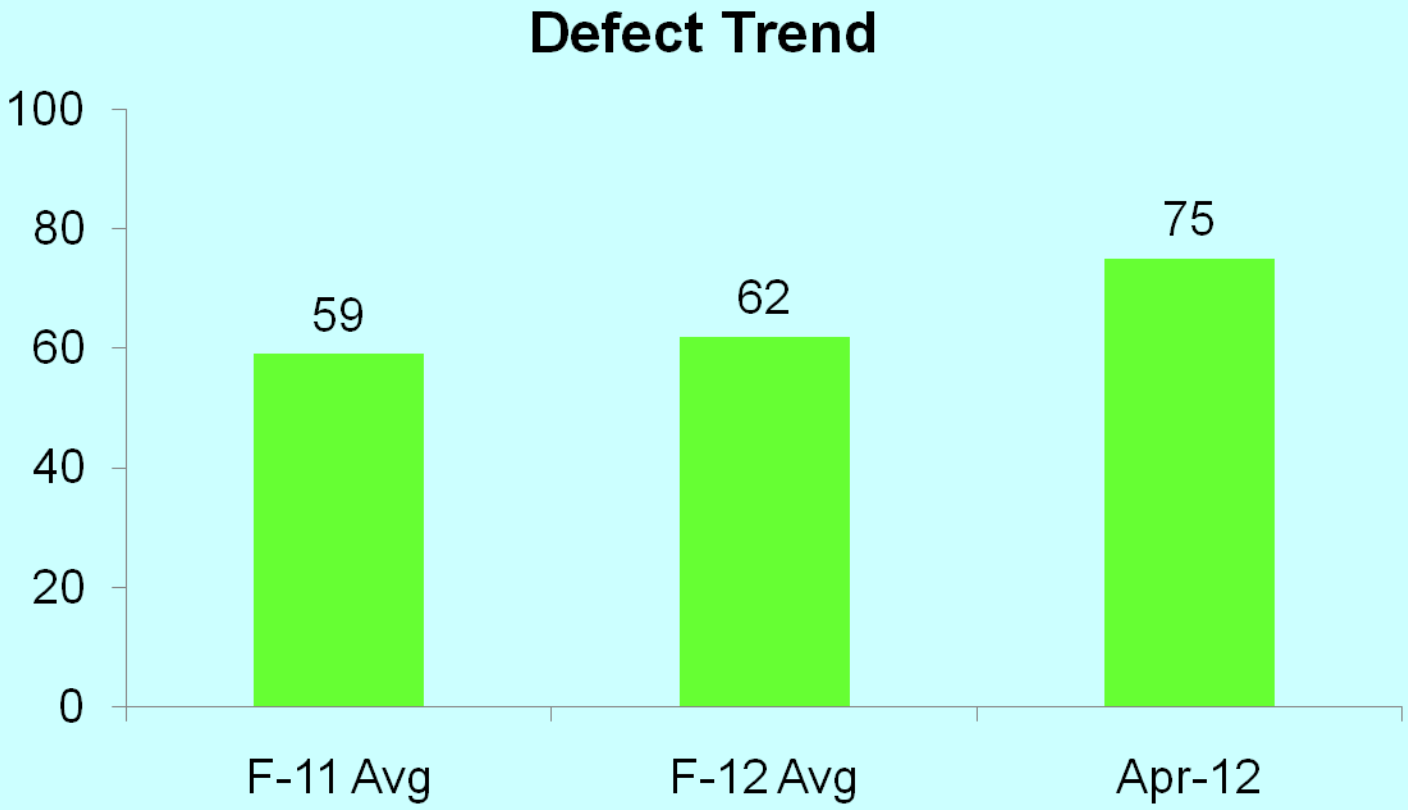
After the process of clinching assly moves to # 50

3. Understand the current situation

Back Door assly



3. Understand the current situation



4. Select Target

Present Level:

62 nos for last average per month

Target:

ZERO per month

TEST OF HYPOTHESIS

Brainstorming By CFT

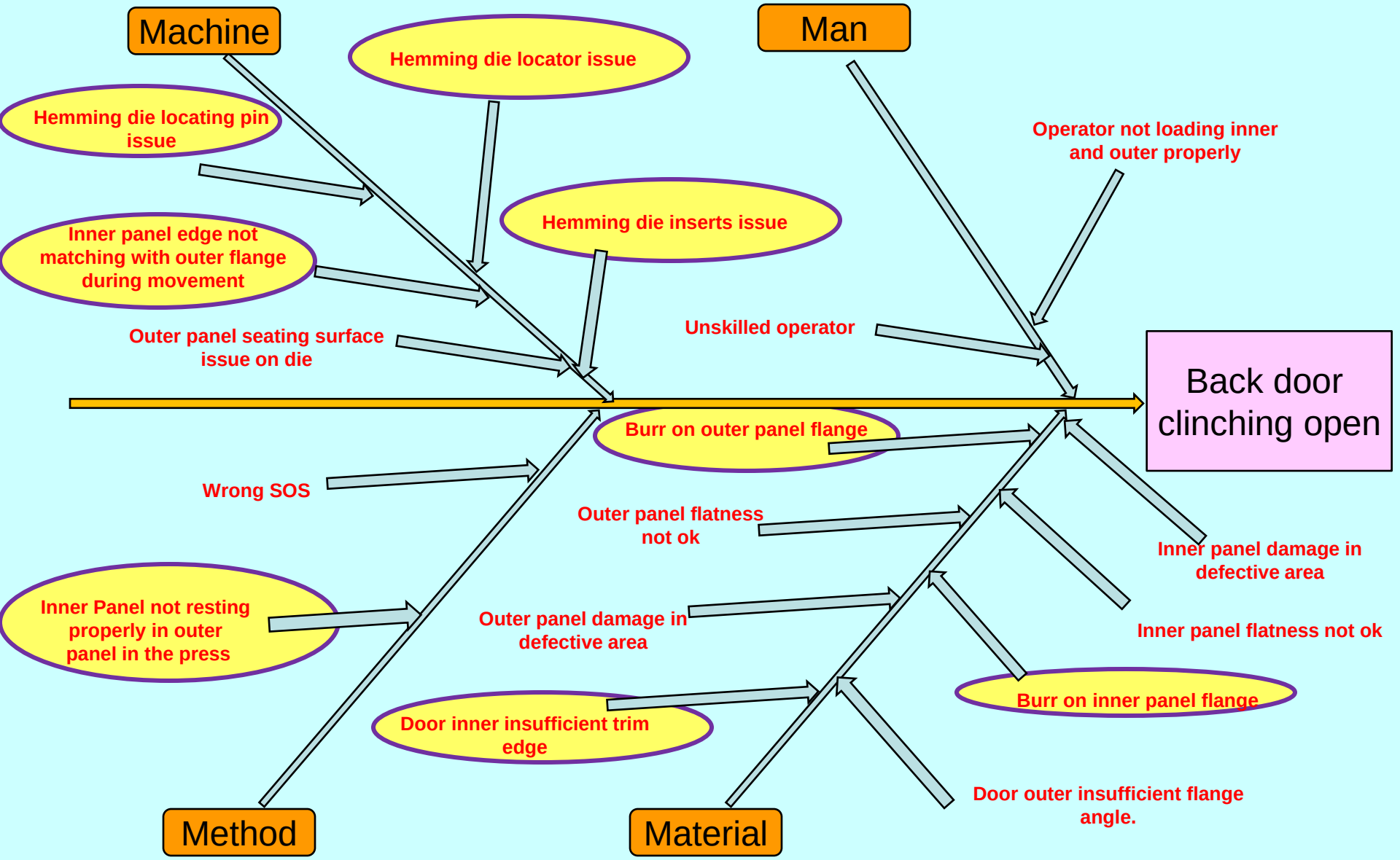
Sr. No	Possible Causes
1	Hemming die locating pin issue
2	Outer panel seating surface issue on die
3	Hemming die locator issue
4	Hemming die inserts issue
5	Burr on outer panel flange
6	Outer panel flatness not ok
7	Outer panel damage in defective area
8	Door inner insufficient trim edge
9	Inner panel damage in defective area
10	Inner panel flatness not ok
11	Burr on inner panel flange
12	Door outer insufficient flange angle.

TEST OF HYPOTHESIS

Brainstorming By CFT

Sr. No	Possible Causes
13	Unskilled operator
14	Operator not loading inner & outer panel properly
15	Inner panel not resting in outer panel in the press
16	Wrong SOS.
17	Inner panel edge not matching with outer panel flange during movement

5.1 Identify possible causes:



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Cause
1	Hemming die locating pin issue
2	Hemming die locator issue
3	Hemming die inserts issue
4	Burr on outer panel flange
5	Door inner insufficient trim edge
6	Burr on inner panel flange
7	Inner panel not resting in outer panel in the press
8	Inner panel edge not matching with outer panel flange during movement

TEST OF HYPOTHESIS

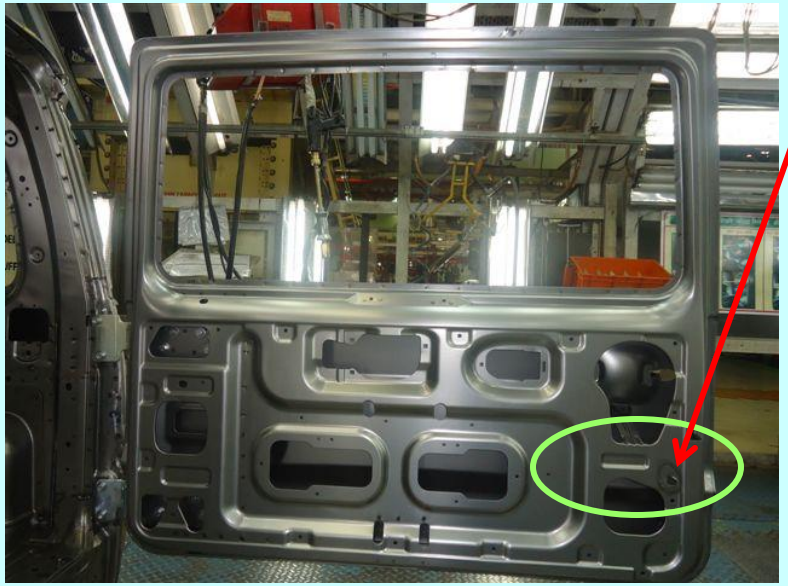
5.2 Confirm the probable cause:

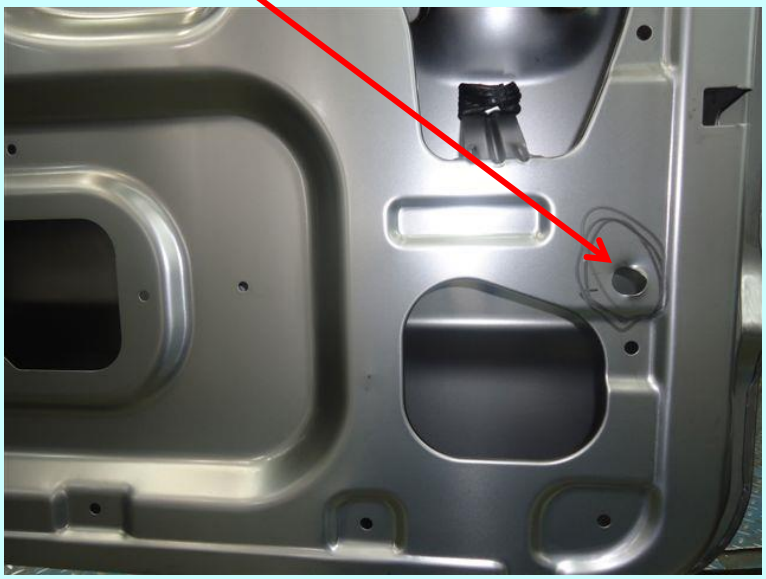
Sr. No	Probable Causes	Testing and Observations	Conclusion
1	Hemming die locating pin issue	Locating hole of the door getting damaged	Hypothesis Valid
2	Hemming die locator issue	Found worn out at defective area	Hypothesis Valid
3	Hemming die inserts issue	Found worn out at defective area	Hypothesis Valid
4	Burr on outer panel flange	Burr observed more than 10% of material thickness	Hypothesis Valid
5	Door inner insufficient trim edge	Found ok on panel checker	Hypothesis Invalid
6	Burr on inner panel flange	Burr observed more than 10% of material thickness	Hypothesis Invalid
7	Inner panel not resting in outer panel in the press	Inner panel edge rests on outer panel flange in defective area.	Hypothesis Valid
8	Inner panel edge not matching with outer panel flange during movement	Sagging of outer panel during movement on loader	Hypothesis Valid

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Causes	Testing and Observations	Conclusion
1	Hemming die locating pin issue	During process walk through, Locating hole of the door found damaged in 5 doors & same doors are reported for the clinching open concern at buy-off.	Hypothesis Valid



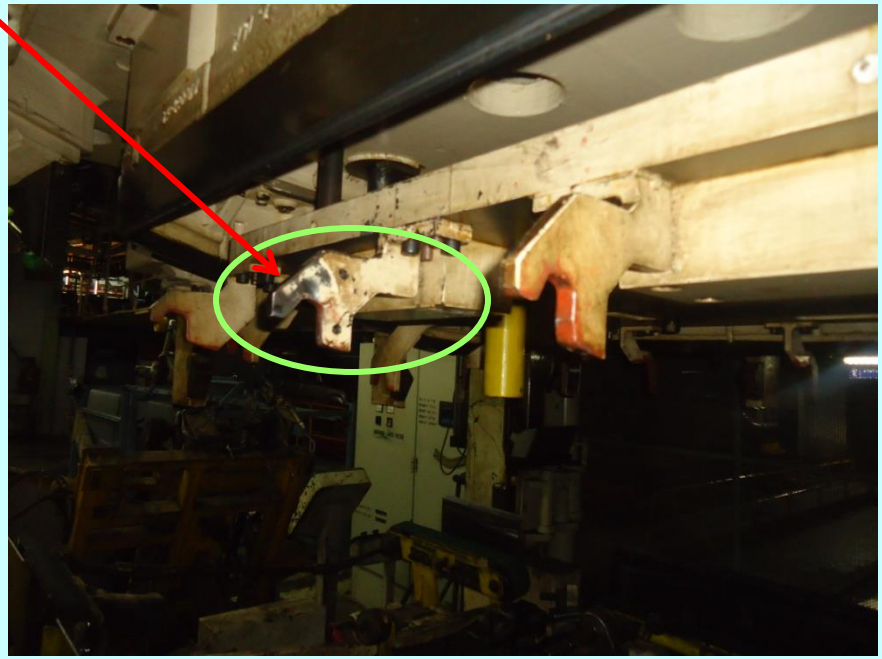


TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Causes	Testing and Observations	Conclusion
2	Hemming die locator issue	Die locator checked found in worn out condition at defective area	Hypothesis Valid

Locator worn out

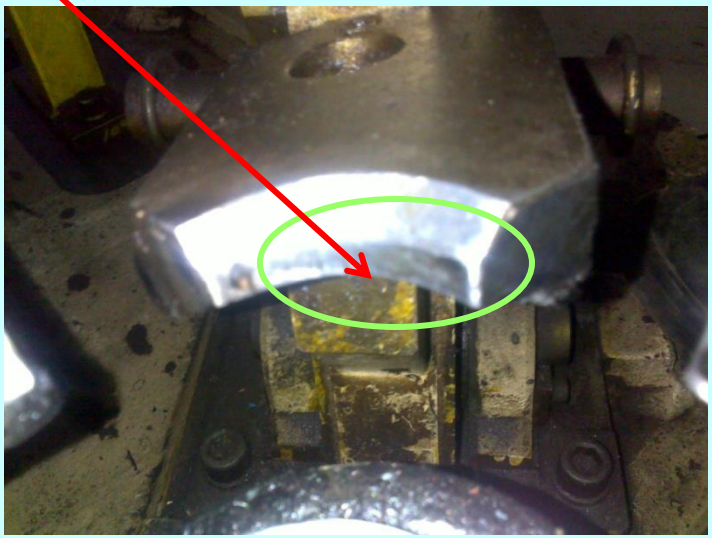


TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Causes	Testing and Observations	Conclusion
3	Hemming die inserts issue	Die Insert found worn out at defective area	Hypothesis Valid

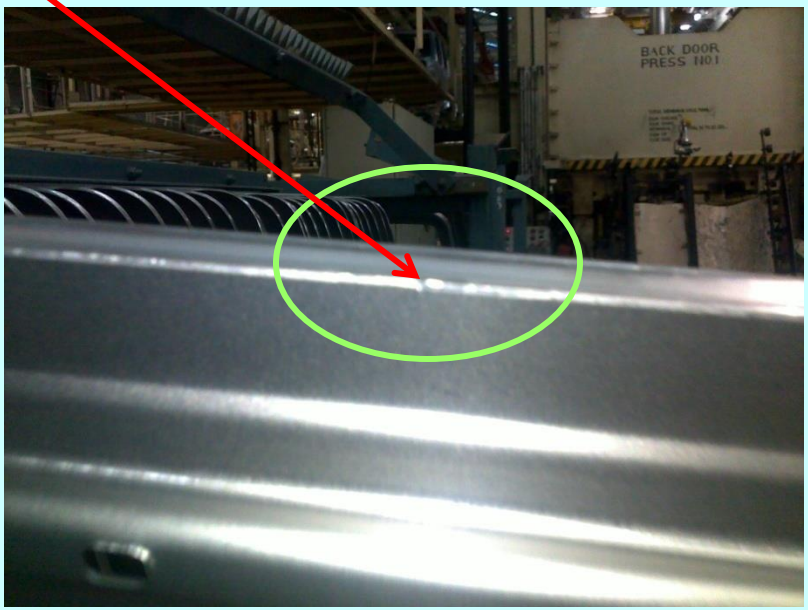
Insert found worn out



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Causes	Testing and Observations	Conclusion
4	Burr on outer panel flange	Press shop panel checked for burr at defective area ,observed more than 10% of material thickness.	Hypothesis Valid



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Causes	Testing and Observations	Conclusion
7	Inner panel not resting in outer panel in the press	Inner panel edge rests on outer panel flange in defective area.	Hypothesis Valid

Inner panel not rests on outer panel during loading

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Causes	Testing and Observations	Conclusion
8	Inner panel edge not matching with outer panel flange during movement	Sagging of outer panel during movement on stn#20 loader.	Hypothesis Valid

WHY - WHY Analysis

5.3 Identify the root causes:

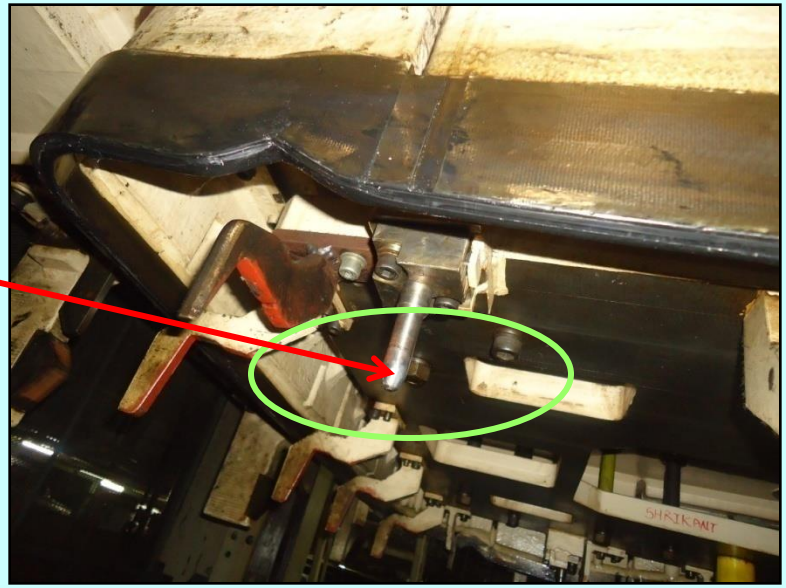
Probable cause	why	why	why	why	Action
Hemming die locating pin issue	Pin entry difficult	Insufficient chamfer to pin	Poor pin design		
Hemming die locator issue	Hemming die locator worn out	Deterioration of locator			
Hemming die inserts issue	Hemming die inserts worn out	Deterioration of inserts			
Burr on outer panel flange	Excessive clearance in cutting edges	Deterioration of inserts			
Inner panel not resting in outer panel in the press	Sagging of outer door due to gravity	Less support to the both door assly	Poor Design of belt support		
Inner panel edge not matching with outer panel flange during movement	Belt support during the movement is less	Sagging of outer door due to gravity	Minimum resting surface of both door assly on belt	Uneven profile of both doors	

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Hemming die locating pin issue	locating pin of 8mm chamfer as per design provided.	17.05.12	Gajanan	Closed

New locating pin provided

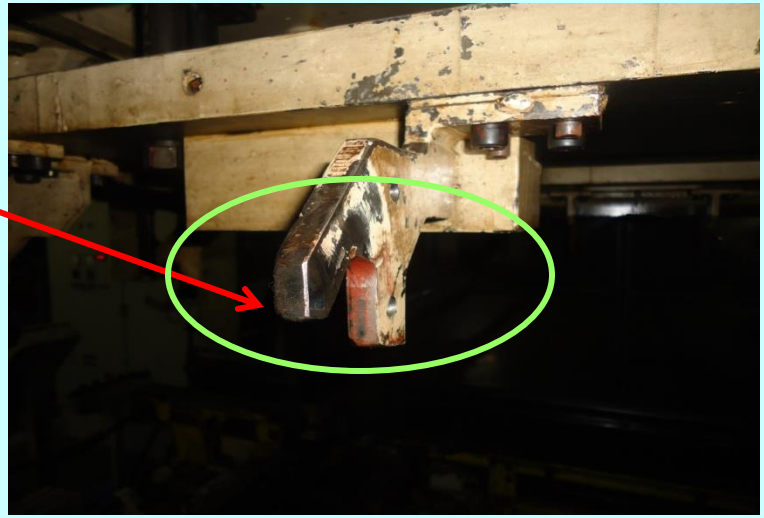


6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Hemming die locator issue	Locator welding & matching done with the matching panel profile.	20.05.12	Gajanan	Closed

Locator matching done



6. Implement the corrective measures

6.3 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Hemming die inserts issue	Die insert welding & matching done with the matching panel profile.	16.06.12	Gajanan	Closed

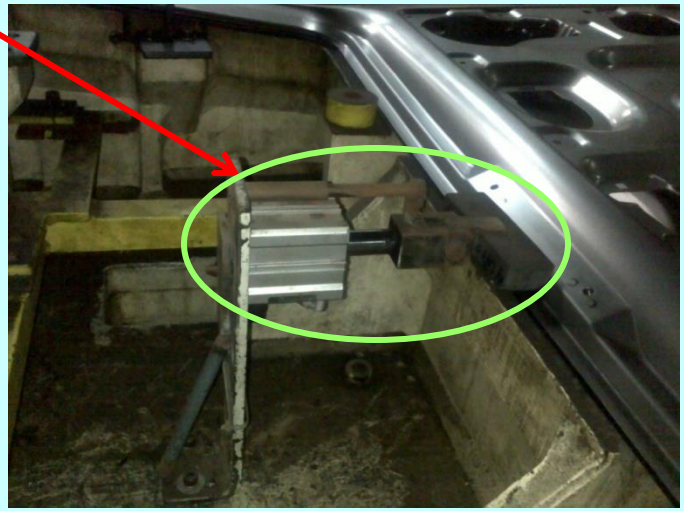
Before

After

6. Implement the corrective measures

6.5 Actions Decided

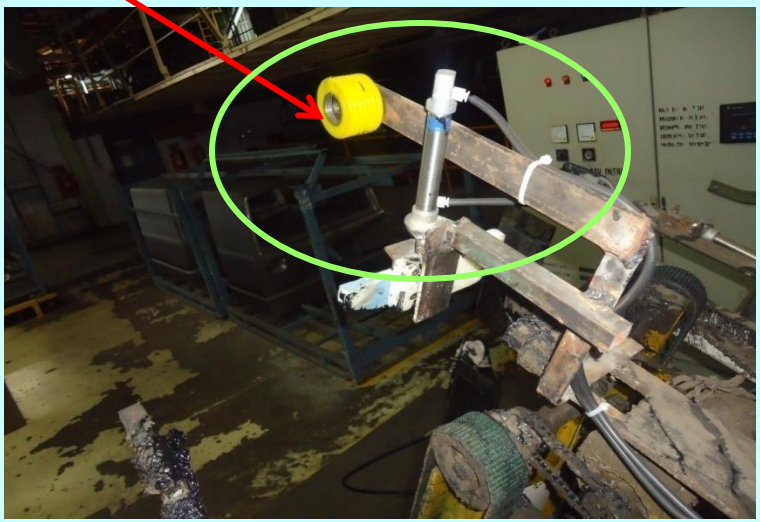
Root Cause	Action Plan	Target Dt	Resp.	Status
Burr on outer panel flange	Die correction done on cutting edge of die inserts	11.07.12	Nimba Patil	Closed
Inner panel not resting in outer panel in the press	New push cylinder is mounted on backdoor stn.30 for proper resting of inner panel in outer panel.	14.08.12	Gajanan	Closed



6. Implement the corrective measures

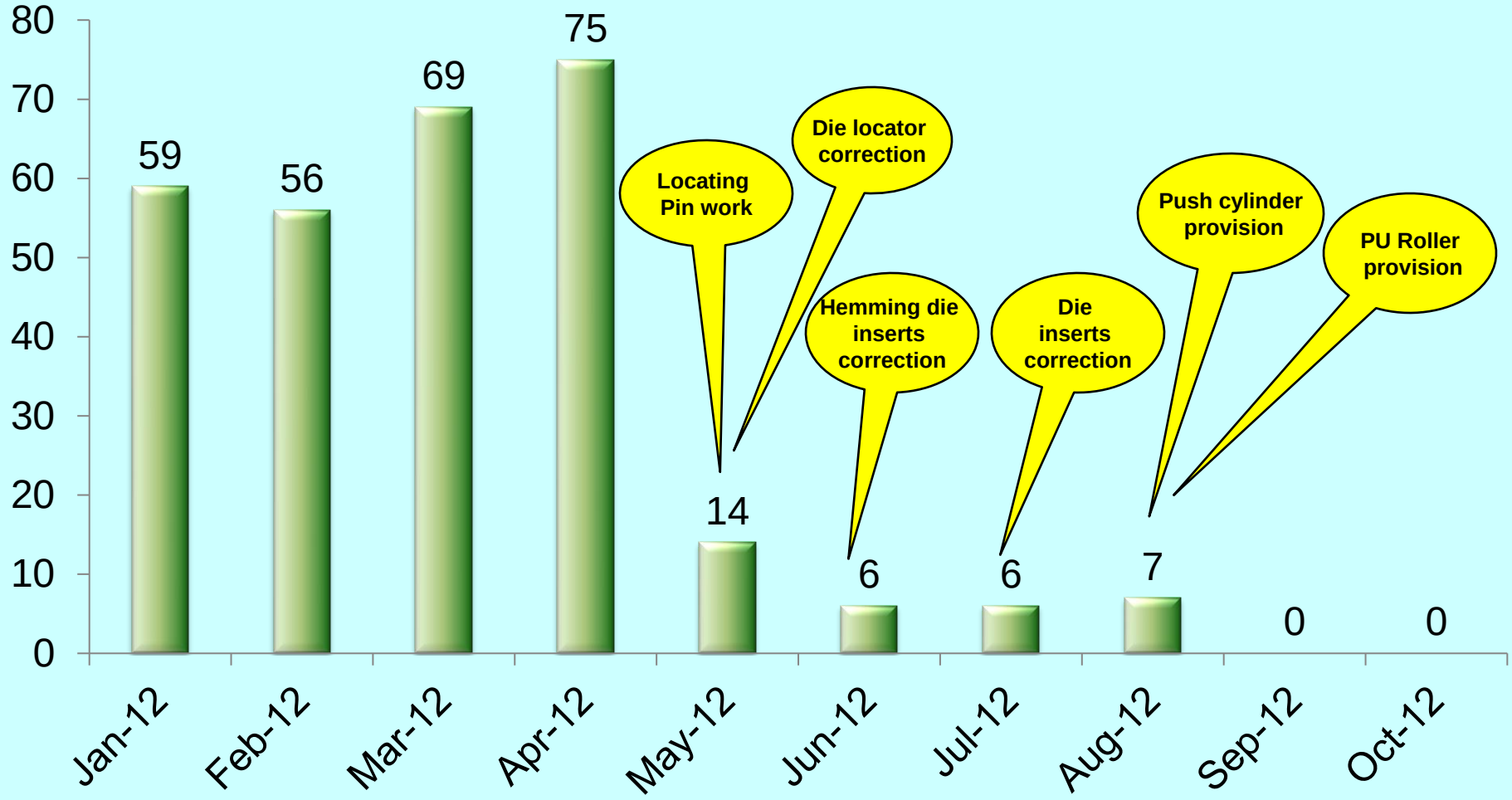
6.6 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Inner panel not resting in outer panel in the press during movement	PU roller is mounted on stn.# 20 to support the inner & outer panel assly to avoid sagging.	25.08.12	Gajanan	Closed



7. Check & Confirm the results & effects

7.1 Verify Results:



Check & confirm the results & effects (Cont……)

7.2 Full Scale Implementation plan:

1) All Decided actions are completed

7.3 Tangible benefits:

1) Annual cost saving due to defect elimination is Rs1699*12 is = **Rs 20393 /Year**
2) FRC is improved by 0.8%
3) Rework time of the operator eliminated by 20 min/body
4) Heavy rework gets eliminated
5) High customer satisfaction through customer voice

Check & confirm the results & effects (Cont……)

7.3 Intangible benefits:

1. Team Morale improved
2. High Customer satisfaction
3. Team Work Built up.
4. Team Confidence Boost up.

8. Standardisation

8.1 Updation of documents:

1.Fixture history card updation for backdoor stn. 20 & stn .30

FIXTURE HISTORY CARD

ZONE: Z20 DEPT : MFG ENGG.-SCORPIO B

FIXTURE NO. Z20-20 FREQUENCY : 6 Months and/or whenever changes done

ASSY NAME: Assy Back door 20

DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
25.08.2012			Backdoor clinching open due to mismatch of inner & outer panel	PU support roller is provided	

FIXTURE HISTORY CARD

ZONE: Z20 DEPT : MFG ENGG.-SCOR

FIXTURE NO. Z20-30 FREQUENCY : 6 Months and/or whenever changes done

ASSY NAME: Assy Back door 30

DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
9/12/2007			Clinching Problem observed	Locator provided for back door bottom area opposite to hinge side.	
27/01/08			Clinching Problem observed	Welding done to preheming die block to improve contact area of block.	
17/06/08			Clinching opening observed at opposide side of hinge due to pre-hemming die mechanism casting block broken	Pre-hemming die mechanism casting block welded by adding support plate.	
15/10/2008			Pre-hemming die mechanism casting block broken	Pre-hemming die mechanism casting block welded by adding support plate.	
6/11/10			Heavy clinching open out side ways due to profile disturb	Prehemming block profile matched by spotting	
5/3/11			Clinching not ok back door top hinge area	Prehemming block die spotting done	CA3P0087
17/5/2012			Loacting hole damage observed result in clinching open	round locating pin chamfering done	
20/5/2012			Back door clinching open	upper die locator welding & matching done	
16/6/2012				Lower die insert welding & matching done	
5\8\12			Clinching open at upper side	Die locator 0.2 shim added	CA2P623
6\8\12				Location of locator in +ve Z shifted by 0.5mm	
14/8/2012			Clinching open conern due to inner edge resting on flange of outer door	push cylinder unit is provided	

8. Standardisation

8.2 Training:

OPL given to all stage operators.

8.3 Additional Audit / Check

1) Internal Customer Audit.
2) IQ Audit.

9. Summarise & Plan future actions

9.1 Conclusion:

An innovative approach and team work by doing GEMBA study can lead to elimination of such chronic concerns

9.2 Horizontal Deployment:

New chronic concern initiated as Front door Lh clinching opening in Scorpio

9.3 Lessons Learnt:

Things Gone Wrong :

Less support given to both the door assly during design belt conveyor which due to that back door assly not matched properly & defect generated.

Things Gone Right :

We study the process of clinching thoroughly & probable root causes validated in proper way

Thank You……MQS TEAM